

JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science & Artificial Intelligence

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PROFESSIONAL PROFILE

9th-semester (final) Computer Science & AI student at Universidad Sergio Arboleda, focused on data science, machine learning, and AI systems. Hands-on experience with data pipelines, predictive models (XGBoost, LightGBM, neural networks), cloud infrastructure (AWS), real-time-constrained distributed systems, and production AI applications — both in independent projects and for real clients. I take initiative to push projects beyond the classroom, work well in teams, and stay oriented toward concrete results.

EDUCATION

Universidad Sergio Arboleda — Bogotá, Colombia

2021 – Present

B.Sc. Computer Science & Artificial Intelligence | 9th semester of 9

GPA: 4.04/5.0 (Colombian scale) • German equivalent ≈ 2.4 (Modified Bavarian Formula) • Upward trend: last 3 semesters avg. 4.5+/5.0

Completed coursework: Big Data & Data Engineering, Machine Learning, Simulation & Modeling, Neural Networks (Classification, Elective), Engineering Communication, Ethics & Social Responsibility.

FEATURED PROJECTS

NEXUS — Autonomous Personal AI System

2025 – Present

Python 3.11 • 5-tier LLM routing (Cerebras/NVIDIA NIM) • GCP hybrid deploy • Custom wake word • Cloned voice (ElevenLabs) • MCP server
Independent project, closed-source by design: the public repo documents the architecture, not the code.

- Architected 5-tier LLM routing (Ollama → Groq → Cerebras 2600 tok/s → NVIDIA NIM 128K-context multimodal → Gemini fallback) with 38 auto-discovered skills spanning home automation, personal finance parsing, and passive security monitoring.
- Built an MCP server exposing system capabilities as tools for external AI agents; deployed a hybrid cloud/local architecture on a free-tier GCP VM (Tailscale-tunneled, service-account impersonation) running core services 24/7 with automated cost-guard cutoff.
- Designed safety-first automation: confirmation-gated system power control and allowlisted actions ensure irreversible commands never fire on a single ambiguous phrase; custom-trained wake word (ONNX) and cloned voice via ElevenLabs with local fallback.
- Vector memory via ChromaDB for context persistence; real home automation (IoT) and personal finance analysis active in production.

Victor — Voice-Driven AI Race Engineer for Sim Racing

2026

Independent project: a native-Linux alternative to CrewChief, with Spanish voice responses generated by an LLM.

Python • C • Wine/Proton • Groq/Cerebras/NVIDIA NIM/Gemini • Piper TTS • pytest

- C shared-memory reader running under the same Wine/Proton prefix as the game to read Assetto Corsa's real telemetry (the same source CrewChief uses).
- Chained failover across 4 LLM providers (Groq → Cerebras → NVIDIA NIM → Gemini) with a per-provider circuit breaker, for resilience during a live race session.
- Deterministic, no-LLM spotter (150ms debounce) for side-by-side car alerts; Spanish speech recognition and synthesis running natively on Linux, with no Windows Speech Recognition anywhere in the pipeline.
- 143 pytest tests, CI green on Python 3.10 and 3.12; technical case study documenting the architecture decisions and their trade-offs.

SAVI v2 — Autonomous Real Estate Valuation System

2026

Team project (with Valeria Larea, Nicolás Garzón, and Juan Niño) for the Machine Learning course.

Python • XGBoost • K-Means • MDP • Q-Learning • Deep Q-Network • PyTorch • scikit-learn

- Architected full RL decision pipeline: MDP Value Iteration (convergence 259 iter, $\gamma=0.95$) → Q-Learning tabular (8,000 episodes, ϵ -greedy decay) → Double DQN (PyTorch, 150 epochs, replay buffer 20K, Adam lr=3e-4).
- Consensus policy across all three agents autonomously APPROVE/REVIEW/REJECTs valuations based on expected financial risk; XGBoost + cluster feature achieves $R^2=0.9609$ on 20,203 properties (2006–2024).
- Delivered an IEEE-format technical paper, an interactive HTML demo on GitHub Pages, and a full reproducible Python pipeline.

Ayuda Terremoto Colombia — Distributed Humanitarian Coordination

2026

Simulated scenario (a hypothetical earthquake in Chocó, Aug. 2026) used as a realistic design driver; not a deployment responding to an actual event.

Python • FastAPI • PostgreSQL • React • Vite • IndexedDB • Render

- Hybrid architecture: FastAPI/PostgreSQL backend as the national node + an offline-first local node (React/Vite + IndexedDB) with a per-action retry sync queue.
- Report deduplication via fuzzy matching (no LLM, a deliberate choice for auditability); automatic activation via USGS polling and HXL-format export.
- 80+ pytest tests on the backend and ~650 lines of Vitest tests on the local node; deployed on Render (backend + public portal).

- Reinforcement learning: Q-learning (FrozenLake), REINFORCE (CartPole), DQN with replay buffer.
- CNNs: manual convolution and a SmallCNN (dropout, weight decay) trained on CIFAR-10.
- Complex systems simulation: forest-fire spread with firefighter agents (FSM, incendios_agentes.py) and a SEIR+H epidemiological model with RK4/Euler numerical integration (epidemia_dinamicos.py).

ShopStream — End-to-End AWS Big Data Pipeline2026

Python · PySpark · AWS Lambda · AWS EMR · AWS Glue · RDS PostgreSQL · Flask · Zappa

- Engineered an end-to-end AWS pipeline for an e-commerce platform: 2.5M synthetic events generated → partitioned S3 ingestion → Lambda validation (CloudWatch metrics + quarantine) → EMR/PySpark analytics (6 KPIs + anomaly detection) → Glue ETL → RDS PostgreSQL DWH → Flask API via Zappa.
- 86% test coverage with pytest; CI/CD via GitHub Actions; hardened RDS security groups for controlled access.

NLP API — spaCy + FastAPI on AWS2026

Team project (4 members) for the Natural Language Processing course.

Python · spaCy · FastAPI · AWS Lambda · AWS EC2 · Docker · pytest

- spaCy NLP pipeline (POS tagging, lemmatization, NER, dependency parsing) plus TF-IDF/BoW/one-hot vectorizers implemented from scratch to meet a non-standard spec.
- Deployed to AWS Lambda (Function URL) and EC2 (AWS Academy); CI running pytest and a Docker build.

Project Dogma — Stochastic Social Propagation Engine2026

Team project led by a classmate; no public repo (code owned by the team lead).

TypeScript · React

- Ideological-propagation simulator built on Markov chains, hidden Markov models, and game theory. Presented at Data Fest — Universidad Sergio Arboleda (Rulo Científico track); [instagram.com/p/DYU3Q1BIUdn](https://www.instagram.com/p/DYU3Q1BIUdn)

Big Data Platform — AWS (Chinook / Sakila / Credit)2026

Chinook: team project (with Juan Hurtado and David Martinez) for the Big Data course.

AWS EC2 · RDS (PostgreSQL / MySQL) · Athena · S3 · AWS Glue · Terraform · React · FastAPI · Git

- Fullstack on Chinook DB: React/Vite frontend + FastAPI backend on two EC2 instances, PostgreSQL on RDS; 17-section technical document.
- Loaded Sakila DB into RDS MySQL, resolving the log_bin_trust_function_creators restriction via a Custom Parameter Group.
- Hive-partitioned external table (credit2) on Athena + S3, resolving a case-sensitivity issue in partitions via manual ALTER TABLE.
- NYC Yellow Taxi dataset (94M rows, 12 months) on AWS EMR with PySpark; resolved a schema inconsistency across Parquet files using explicit StructType and per-file normalization with unionByName.

Custom Programming Language2023

Python · ANTLR4 · Tokenization · Interpretation

- Designed and implemented a Python-inspired language with a custom tokenizer, lexer, and parser, and a visitor-pattern execution engine, using ANTLR4.

FREELANCE / CLIENT WORK

Zafra Smart CRM — real client (Zafra Asesores Tributarios)2026

Freelance engagement for a tax-advisory firm in Girardot, Colombia; built and demoed to the client.

Python · FastAPI · SQLAlchemy · React/Vite · Groq (Llama 3.3 70B) · WhatsApp Cloud API

- FastAPI/SQLAlchemy backend with a full relational model (clients, prospects, campaigns, interactions) and a React/Vite frontend for prospect-pipeline management.
- AI layer using Groq/Llama 3.3 70B for summaries, email drafting, and lead scoring; a qualification bot integrated with the WhatsApp Cloud API.
- Technical documentation delivered to the client; auth and continuous deployment remain open items before production use.

ReushiGo — ordering platform for a family business2026

Full-stack tool for a family coffee-distribution business; private repository.

React · Vite · Tailwind · Express · Prisma · Telegram Bot API

- Admin panel (React/Vite/Tailwind) and backend (Express/Prisma) with admin-key authentication and rate limiting.
- Telegram sales bot with interactive menus and natural-language ordering; a real 19-product catalog.
- GitHub Actions CI (lint, build, npm audit, tests) and its own security-review document.

TECHNICAL SKILLS

Programming Languages	Python (intermediate-advanced), C, Bash Scripting, JavaScript (basic), SQL
ML / AI	scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, spaCy, Whisper, Ollama

Big Data & Cloud	AWS (EC2, RDS, S3, Athena, Glue, EMR), Terraform, Git / GitHub, Docker (basic)
Data & ETL	Pandas, NumPy, Matplotlib, openpyxl, Power BI, Jupyter Notebook, PySpark
Automation	Playwright, pywinauto, Telegram Bot API, Gmail / Calendar OAuth2
LLM APIs	Groq, Cerebras, NVIDIA NIM, Google Gemini, ElevenLabs
Backend / Web	FastAPI, Flask, Express, React, SQLAlchemy, Prisma
Other	ANTLR, CUDA, NetworkX, ChromaDB, ZeroMQ, tkinter, pytest

ACHIEVEMENTS & RECOGNITION

- 1st Place — International Competition Wisibilízalas (WarmiTics)

2019

Universidad Pompeu Fabra · Barcelona, Spain · Shakespeare School, Quito, Ecuador · Senior Category

 - 1st place in an international competition organized by Universidad Pompeu Fabra, representing Shakespeare School (Ecuador) in the Senior category at age 17, against teams from Latin America and Europe.
 - As Technical Product Manager and Lead Frontend/UX, architected and developed the full web application; coordinated cross-functional logistics; conducted structured interviews with women leaders in technology to build the site's primary content.
 - Winning project available at: sites.google.com/view/warmitics
- 1st Place — USABOT Robotics Challenge

 - Led hardware design and go-to-market strategy for a smart automatic irrigation system, winning 1st place against competing university teams. Designed the 3D CAD model for the physical prototype, integrated ultrasonic sensor architecture for real-time water-level detection and automated valve control, and wrote and delivered the pitch that won over the judges.

CERTIFICATIONS

Database Design — Oracle Academy — August 2024

Database Programming with SQL — Oracle Academy — December 2024

AWS Academy Graduate — Data Engineering — Amazon Web Services Academy — May 2026 — 40 hours · Credly: credly.com/go/UGp1CUEB

LANGUAGES

Spanish: Native • English: B2+ (technical and academic use: documentation, presentations, reports)

SOFT SKILLS

- Technical leadership: led CAD design and sensor architecture for USABOT (1st place), including the pitch delivered to judges.
- Effective communication: interviewed women leaders in technology for WarmiTics, translating their stories into content for an international audience (1st place, Wisibilízalas).
- Teamwork: coordinated with 3-to-4-person teams on SAVI v2 and the Big Data Platform (AWS), both collaboratively built course projects.
- Analytical thinking: designed the multi-agent consensus policy (MDP + Q-Learning + DQN) behind SAVI v2 for financial risk decisions.
- Self-directed learning: built NEXUS, a personal AI system, entirely outside the academic curriculum.
- Commitment and resilience: sustained NEXUS and other independent projects alongside a full course load, with an upward-trending GPA (last 3 semesters above 4.5/5.0).

Available for internships, freelance projects, or collaborations in AI, Data Science, and Software Development.